A Dual-Modal Approach Using Electromyography and Sonomyography Improves Prediction of Dynamic Ankle Movement: A Case Study

Qiang Zhang, *Student Member, IEEE*, Ashwin Iyer, Ziyue Sun, Kang Kim, *Senior Member, IEEE*, and Nitin Sharma, *Senior Member, IEEE*

***Abstract*— For decades, surface electromyography (sEMG) has been a popular non-invasive bio-sensing technology for predicting human joint motion. However, cross-talk, interference from adjacent muscles, and its inability to measure deeply located muscles limit its performance in predicting joint motion. Recently, ultrasound (US) imaging has been proposed as an alternative non-invasive technology to predict joint movement due to its high signal-to-noise ratio, direct visualization of targeted tissue, and ability to access deep-seated muscles. This paper proposes a dual-modal approach that combines US imaging and sEMG for predicting volitional dynamic ankle dorsiflexion movement. Three feature sets: 1) a uni-modal set with four sEMG features, 2) a uni-modal set with four US imaging features, and 3) a dual-modal set with four dominant sEMG and US imaging features, together with measured ankle dorsiflexion angles, were used to train multiple machine learning regression models. The experimental results from a seated posture and five walking trials at different speeds, ranging from 0.50 m/s to 1.50 m/s, showed that the dual-modal set significantly reduced the prediction root mean square errors (RMSEs). Compared to the uni-modal sEMG feature set, the dual-modal set reduced RMSEs by up to 47.84% for the seated posture and up to 77.72% for the walking trials. Similarly, when compared to the US imaging feature set, the dual-modal set reduced RMSEs by up to 53.95% for the seated posture and up to 58.39% for the walking trials. The findings show that potentially the dual-modal sensing approach can be used as a superior sensing modality to predict human intent of a continuous motion and implemented for volitional control of clinical rehabilitative and assistive devices.**

***Index Terms*— B-mode ultrasound imaging, surface electromyography, machine learning regression, dynamic ankle dorsiflexion motion, human limb intent.**

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I. Introduction

WEAKENED function or dysfunction of the tibialis anterior (TA) muscle, which is the primary contributing muscle for ankle dorsiflexion, severely impedes normal walking and balance functions. Neuromuscular disorders due to spinal cord injury or stroke often cause its weakening or dysfunction, hindering normal gait function, particularly ground clearance during the swing phase [1].

Recent assistive and rehabilitative devices such as powered ankle exoskeletons [2], [3] or functional electrical stimulation (FES) [4], [5] can improve weakened ankle dorsiflexion function. To maximize the benefits of these devices, a user must be actively involved in rehabilitation training. Human-in-the-loop control strategies encourage active involvement by providing assistance only when needed [6], [7]. These strategies, thus, require accurate measurements of residual volitional intent to generate stable and effective assistance. Force or torque sensors installed on an exoskeletal frame can be used to measure human limb motion intent. However, feedback from these sensors is reactive, i.e., a limb movement must precede for the sensor to register the applied force or torque. Thus, the assistance, in this case, may not be synchronized with the user. In addition, measurement inaccuracies can creep in because the exoskeletal frames are prone to misalignment between human joint and machine hinge rotation centers, introducing undesired interaction force [8], [9]. In severely weakened muscles, where it is often difficult to produce limb movements, predicting the user's volitional intent with

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these sensors would be challenging. Therefore, instead of a reactionary method, a predictive sensing methodology would increase the assistive device's synchronization and stability.

Surface electromyography (sEMG) uses non-invasive electrodes, attached to the skin, to measure electrical potentials due to the excitation of superficial muscles motor units. sEMG signals' amplitude and frequency positively relate to muscle activation levels [10], [11]. Even in severely weak muscles, it can read the intent despite no observable limb motion [12]–[15]. Thus, biomedical engineers usually use sEMG as a control signal for rehabilitative devices such as prosthetics and exoskeletons on upper and lower limbs, or intelligent wheelchairs. These devices may deploy sEMG-driven models such as a neuromusculoskeletal model (HNM) [16], [17], support vector machine (SVM) model [18], and artificial neural networks [19] to estimate limb joint angles. However, shortcomings of sEMG, such as interference or cross-talk from adjacent muscles and an inability to measure signals from deeply located muscles [20], [21] may prevent accurate joint motion estimation.

Alternatively, ultrasound (US) signals have been recently introduced as another non-invasive sensing interface to estimate intended joint motion. Due to the US signals' ability to directly measure and distinguish adjacent muscle contractions, even at different depths, US signals-based joint posture or motion estimation may not suffer from the shortcomings of sEMG. Its inherent high spatiotemporal resolution and specificity enabled investigators to successfully infer voluntary finger or wrist movement and dexterous control of robotic hands [22]–[25]. Both one-dimensional US signal (A-mode) and two-dimensional US imaging (B-mode) have been used for joints motion prediction. Compared to the A-mode US signal, the B-mode US imaging-based sensing interface has the advantages of a high signal-to-noise ratio and directly visualizing morphology deformation. Studies in [10], [26]–[29] extracted a variety of morphological and functional features of B-mode US imaging including, pennation angle (PA), fascicle length (FL), muscle thickness, cross-sectional area, echogenicity, tissue displacement or strain, and tissue optical flow, to represent the muscle deformation changes and predict human joints' postures or volitional motion.

The majority of the current studies related to the use of sEMG signals or US signals for the human limb motion estimation focused on upper extremities [22]–[25], [30], [31]. However, a few recent studies focused on continuous knee joint motion estimation [29], [32], and ankle joint motion prediction using sEMG or US signals [33], [34]. These studies mainly used uni-modal bio-signals, i.e., either use features from sEMG signals or US signals, to estimate or predict discrete joint postures or continuous joint motion. A recent review study on interpreting hand motions [35] summarized multiple sensing technologies (like EMG and near-infrared spectroscopy, electroencephalography and electroneurography), known as multi-modal bio-signals, to get a higher limb motion estimation or prediction accuracy than uni-modal bio-signals. Our previous studies also showed a dual-modal approach that combines neuromuscular features from TA's sEMG signals and US imaging [10], [11] improves isometric dorsiflexion moment prediction. The dual approach to combine sEMG signals and US imaging may be beneficial as it captures complementary information of muscle activation levels (electrical aspect) and muscle contractility (mechanical aspect) of the targeted skeletal muscle. However, the dual-modal approach remains challenging and unexplored for the continuous joint motion estimation due to the dynamic characteristics of sEMG and US imaging signals.

In this paper, the objective was to investigate the benefits of a dual-modal approach for predicting volitional ankle dorsiflexion motion under dynamic conditions. The current study is a next logical step from our previous study on isometric ankle motion and brings us closer to ultimately address the practical need to assist ankle joint dorsiflexion for patients with drop-foot. Although the HNM that combines sEMG and US imaging as mentioned in [10], [11] is also implementable in a dynamic scenario, a system identification of muscle-tendon-unit geometry, muscle contraction and activation dynamics, and joint-segment dynamics is needed to realize the joint motion prediction from the neuromuscular signals. Further, the dual-modal approach in [10] determined optimal allocation ratios to combine sEMG and US imaging signals. Alternatively, this paper presents a dual-modal approach based on black-box modeling methods. We hypothesized that the proposed dual-modal approach improves accuracy of the ankle dorsiflexion motion prediction compared to the uni-modal approach, irrespective of the employed machine learning regression models. The tested machine learning regression models in this study include multiple linear regression (MLR), feedforward neural network (FFNN), support vector machine regression (SVR), and Gaussian processing regression (GPR). These black-box approaches circumvent the need to identify multiple system parameters, such as in HNM, and automate the process of combining the sEMG and US imaging signals. The latter advantage enables scaling the number of neuromuscular features as inputs, which may generate a comparative or superior prediction performance compared to an HNM approach. Compared to [10], [11], the main innovations and contributions are:

- extracting multiple sEMG-derived and US imaging-derived explicit features from the TA muscle to characterize the ankle joint dynamic motion both in a seated posture and walking tasks,
- determining the dominant features from the sEMG and US imaging feature sets, where the features were chosen based on the importance weights rankings that were computed through the *RReliefF* algorithm [36],
- investigating the leading time gap of US imaging's dominant features before the upcoming dorsiflexion motion,
- developing a dual-modal approach that combines the dominant features derived from the sEMG signals and US imaging to predict the ankle dorsiflexion motion for two scenarios: the seated posture and walking swing phase, and
- validating the superior continuous prediction performance of the proposed dual-modal approach over uni-modal approaches through multiple machine learning regression models.

This paper is organized as follows. Section II presents the experimental setup for both the seated posture and walking trials at multiple speeds, data collection, features extraction from sEMG signals and US imaging, multiple regression models training and prediction, and statistical analysis. The results of developing uni-modal and dual-modal feature sets, training, and prediction using regression models are presented in Section III, followed by result interpretations and discussions on study limitations and potential implementations. Finally, the conclusion is given in Section IV.

II. Methods

A. Subjects

The study was approved by the Institutional Review Board (IRB) at the North Carolina State University (IRB approval number: 20602), as well as in accordance with the ethical standards of the Helsinki Declaration. An a priori power analysis determined that eight subjects would have 80% power to detect ($p < 0.05$) a difference in the dynamic ankle joint motion prediction by using dual-modal and uni-modal bio-signals feature sets. Therefore, eight young able-bodied participants without any neuromuscular disorders (age: 26.0 ± 2.2 years, height: 173.7 ± 5.9 cm, weight: 72.6 ± 11.2 kg, 2F/6M) were recruited to complete non-weight-bearing dynamic ankle dorsiflexi on movements during a seated posture and dynamic walking experiments on a treadmill at multiple speeds. Every participant signed an informed consent form before participating in the experiments.

B. Experimental Protocol and Apparatus

The experimental setups for the seated posture and walking on a treadmill in this study are illustrated in Fig. 1 (a) and (b). In the first scenario, each participant was seated on a clinical therapy table with adjustable height, and the right foot was kept suspended in the air. The initial right ankle joint posture ($\theta = 0°$) for each dorsiflexion trial was set at the position where the participants felt completely relaxed about their right shank and foot. Each participant was instructed to perform two separate experimental sets during the seated posture. In the first set, the participants were asked and verbally encouraged to volitionally dorsiflex the right ankle joint to the peak toe-up position and stay there for 2 seconds. Three repeated trials were conducted and a 1-minute rest time was provided for the participants to avoid TA muscle fatigue. The averaged position of the recorded 2 seconds data was calculated for each trial. A maximum motion range (MMR) was defined as the highest value among averaged positions.

In the second set, each participant first got a warm-up training to follow a desired sinusoidal dorsiflexion trajectory on the front computer screen for 2 minutes. The trajectory amplitude was between 0° and individual MMR, and the period was set as 2.5 seconds. After the 2-minute warm-up procedure and a 5-minute rest period, the participants were asked to volitionally perform similar dorsiflexion motion without watching the desired trajectory. In the meantime, the sEMG, US imaging, and dorsiflexion kinematics data were collected synchronously. Five repeated trials with each one running for 11 seconds were conducted on each participant. During each trial, the participants were asked to only dorsiflex the right ankle joint while maintaining their right shank static and relaxed. There were 4 complete relaxation-MMR-relaxation cycles within each trial. A 2-minute rest time was provided between two successive trials to avoid the fatigue.

Through physical touch and visual observation of how TA muscle moved during the dynamic ankle dorsiflexion, a targeted region, approximately 30% to 50% of an individual's shank length, distal from the knee joint as shown in Fig. 1 (a), was chosen as a location for sensor placement. The targeted region was cleaned and shaved before attaching the sensors. A similar setup for a differential sEMG sensor (Bagnoli^{TM} Desktop, DELSYS, MA, USA) and a 38-mm clinical linear US transducer (L7.5SC, 6.4 MHz center frequency) were applied in this study as detailed in [10], [11]. The attachment location was selected as 1/3 on the line between the tip of the fibula and the tip of the medial malleolus (distal to the knee rotation center). The US transducer was rotated from a cross-sectional direction to a longitudinal direction and adjusted to obtain the most clearly visualized image for TA muscle. The elevation angle between the holder base (attached to the skin) and holder arm (for stabilizing the US transducer) was designed as a fixed angle of 90°, which guaranteed that the US transducer was always perpendicular to the target muscle area. Conductive and water-soluble US transmission gel was applied to enhance the acoustic coupling and image quality. The depth of US imaging was set as 40 mm to include the entire TA muscle area for all participants.

During both sets of the seated posture, the participant's upper leg was kept at the horizontal position by the therapy table support, and the lower leg was kept almost perpendicular to the upper leg without any knee joint movement. A customized wearable inertial measurement unit (IMU) sensing system was used to measure the ankle joint kinematics. As shown in Fig. 1 (a), 2 low-cost IMUs (MPU 9250, TDK InvenSense Headquarters, CA, USA) were attached on the shank and front foot, respectively. Each IMU includes 3-axis gyroscope and 3-axis accelerometer signals, and the ankle joint angular positions θ were calculated based on a first-order complementary filter (see supplemental section-A). The same setup for the sEMG sensor and US transducer, as mentioned above, was used for the walking experiments at multiple speeds, including 0.50 m/s, 0.75 m/s, 1.00 m/s, 1.25 m/s, and 1.50 m/s. The detailed experimental protocol and apparatus can be found in [37]. After a 1-minute walking warm-up, 20 seconds of data during stable walking were collected for further analysis. During the walking trials, the right ankle joint angular positions were calculated by using Visual 3D software (C-Motion, Rockville, MD, USA) based on the 3D coordinates of 39 retro-reflective markers positioned on the participants' lower extremities.

C. Data Acquisition and Processing

Figure 1 (c) illustrates the flowchart of data acquisition and processing from different sensors. In the seated posture, signals from the sEMG sensor, US machine, and IMUs were

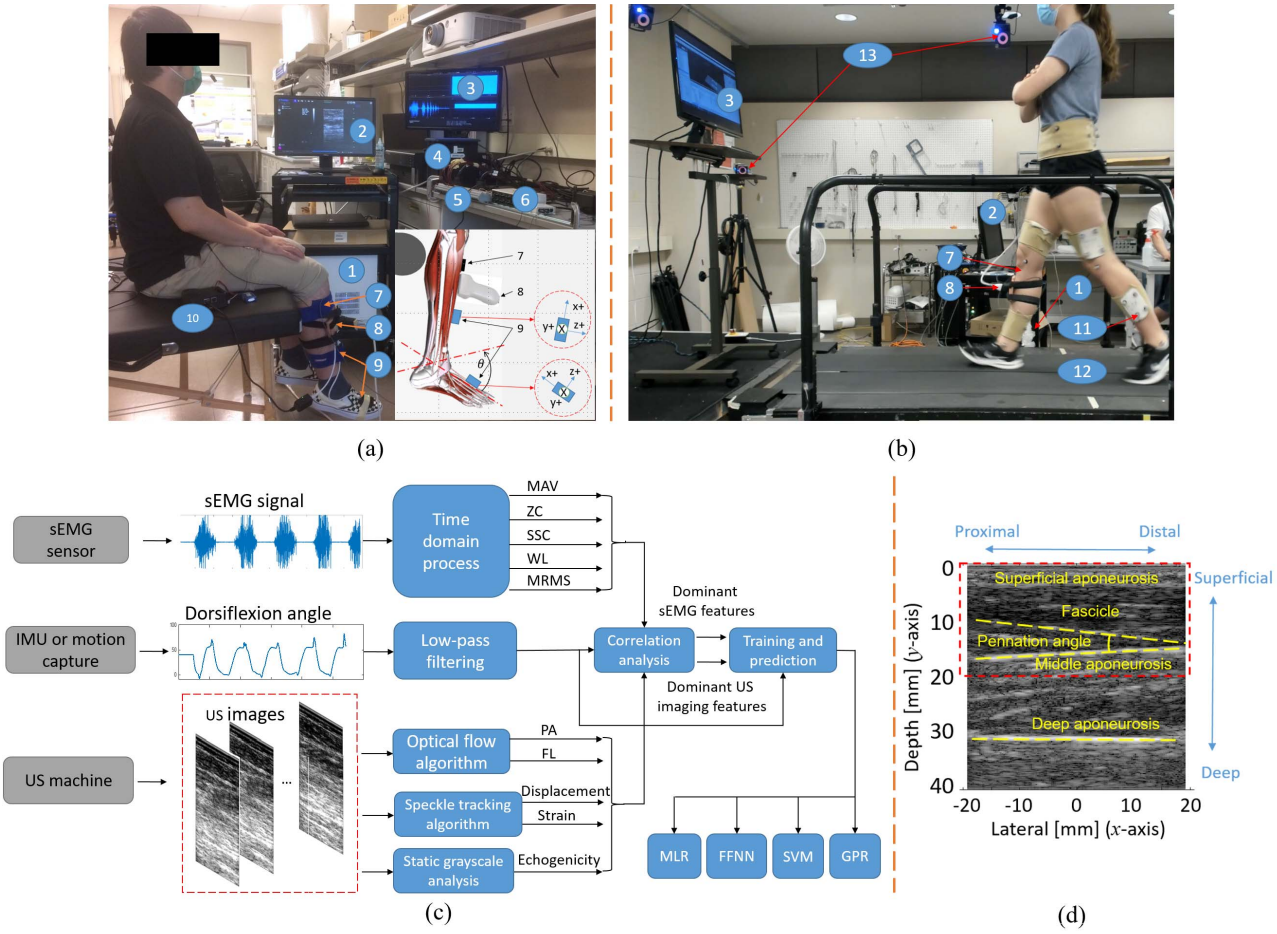


Fig. 1. (a) & (b) The illustration of experimental setups for the seated posture and walking trials at five speeds ranging from 0.50 m/s to 1.50 m/s. 1-US imaging machine. 2-US imaging monitor. 3-Visual feedback screen. 4-Speedgoat target machine. 5-Data acquisition board. 6-sEMG amplifier and filter. 7-Single differential sEMG sensor. 8-Prodigy US transducer. 9-IMU sensors. 10-IMU data acquisition mux. 11-Reflective markers. 12-Instrumented treadmill with two split belts and in-ground force plates. 13-Vicon motion capture cameras. (c) Kinematics and neuromuscular signals acquisition and processing flow, including sEMG and US imaging. Five time domain features from sEMG signal, six morphological and functional features from US imaging were extracted to correlate with the dorsiflexion angle. Three feature sets, including four sEMG features (uni-modal sEMG feature set), four US imaging features (uni-modal US imaging feature set), and four dominant sEMG-US imaging features (dual-modal feature fusion set), were utilized in model training and prediction procedures. (d) A typical US image of TA muscle on Participant A01. Structural features PA and FL are defined in the upper pennate section of the TA muscle. The RGB data of pixels were transferred to gray-scaled values between 0 and 255.

synchronized and sampled at 1000 Hz by using a real-time system programmed in MATLAB/Simulink (R2017a, MathWorks, MA, USA) together with a target machine (Speedgoat Inc., Liebefeld Switzerland). Signals from the sEMG sensor were processed by an input module and a main amplifier (BagnoliTM Desktop, DELSYS, MA, USA) with an amplified gain of 10 k and filtered to the bandwidth between 20 Hz and 450 Hz, finally collected on a 16-bit analog input/output module. Signals from the 2 IMUs entered the collection module IMU Mux by using category 5 cabling, and the IMU Mux wirelessly sent IMUs data packets to a router connected to the target machine. The ultrasound platform was synchronized with the data collection through the use of a 1000 Hz square wave pulse sequence with a duty cycle of 5% as the trigger signal. The US imaging radio frequency (RF) data was saved on the US machine (Prodigy System, S-Sharp, Taiwan) and processed offline. In the walking experiments, signals from sEMG sensor, US machine, and ground reaction force were synchronized and sampled at 1000 Hz while the 3D coordinates of markers were synchronized and sampled at 100 Hz by using Vicon Nexus 2.9 (Vicon Motion Systems Ltd, Los Angeles, CA, USA). Given the ground reaction force, a threshold of 5% body weight was selected to segment a gait cycle into the stance phase and swing phase, and the study focused on the ankle dorsiflexion motion prediction during the swing phase.

According to [38], four classical EMG time-domain (TD) features were extracted, including mean absolute value (MAV), zero crossings (ZC), slope sign changes (SSC), and waveform length (WL). Apart from those, moving root mean square (MRMS) [10] was also extracted. The choice of window length is a balance between sEMG TD features smoothness and signal fidelity. Suitable window length from literature varied from 150 ms [39] to 500 ms [40]. In this study, the window length was selected as 400 ms.

The RF US imaging data stored in the US machine were beamformed to gray-scaled US images offline in MATLAB, as shown in Fig. 1 (d). The brightness and darkness of the US image represent the normalized gray-scaled value (between 0 and 255) of each pixel, which was calculated from a

logarithmically compressed imaging signal. For simplification, only the upper pennate section of the TA muscle was taken into consideration to determine PA and FL as shown in Fig. 1 (d). The region of interest (ROI) was defined as the red dashed rectangular area, where PA was defined as the angle between the most visualized fascicle and the middle aponeurosis. The orientation of the fascicle and the middle aponeurosis with respect to the horizontal line were calculated separately based on optical flow algorithms utilizing an affine transformation to track ending points of the muscle fascicle as determined in the initial US image frame [41], [42]. Then the PA was calculated by summing the two orientation angles, which is denoted as $PA(t_k)$. In addition, the FL could also be calculated as the distance between two ending points, which is denoted as $FL(t_k)$.

Echogenicity, as a functional feature from US B-mode imaging, represents an averaged brightness change of the TA muscle ROI during muscle contraction. It was calculated for the frame at t_k by taking the average gray-scaled value of pixels within the ROI and represented as $Echo(t_k)$. Tissue displacement and strain along with axial and lateral directions were dynamically tracked based on an adaptive speckle tracking algorithm applied in [43], [44]. The US machine configures its axial resolution (0.0385 mm/pixel) much higher than the lateral resolution (0.3 mm/pixel), so the axial strain was preferred and resulted in better smoothing performance for a differentiation filter with a fixed kernel length. The averaged displacement along axial and lateral directions and the averaged strain along axial direction were also calculated and denoted as $D_A(t_k)$, $D_L(t_k)$, and $S_A(t_k)$, respectively, for each image frame within the same ROI defined in Fig. 1 (d). All detailed formulations for neuromuscular features extraction are detailed in the supplemental section-A.

D. Data Training, Prediction, and Statistical Analysis

The importance weights between the sEMG/US imaging neuromuscular feature set and the ankle joint motion from the seated posture were calculated and ranked based on the *RReliefF* algorithm [36]. Then the two features with higher importance weights in the sEMG/US imaging feature set were considered as two dominant sEMG TD features and two dominant US imaging features, respectively. These dominant features can recognize the neuromuscular signals before the upcoming limb motion. Since the leading time of sEMG signal prior to upcoming joint motion has been well investigated in the literature, in the present study, we calculated the leading time's mean value and standard deviation (SD) of the most dominant feature from US imaging (the one with the highest importance weight) prior to the ankle dorsiflexion motion for all eight participants, as shown in Fig. 2. The leading time was determined as the elapsed time period from the first significant change of the most dominant feature (5% of the feature change range from ankle joint relaxation to MMR) and the first significant change of ankle dorsiflexion motion (5% of MMR). The results from these eight participants show that the US imaging leading time varies between 45 ms to 85 ms, which indicates its promising potential to predict an upcoming limb movement.

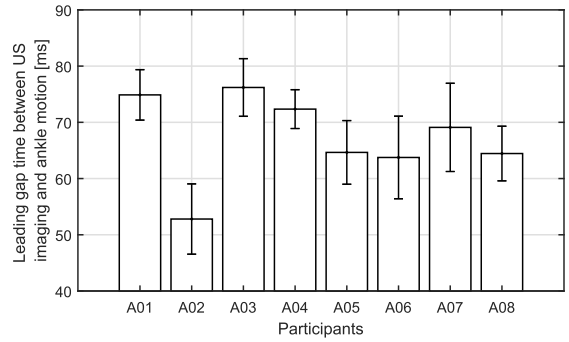


Fig. 2. Results of US imaging leading time gap (mean ± SD) prior to volitional ankle dorsiflexion motion across trials for each participant. Each bar plot consists of 20 data points (4 leading time gaps in each trial × 5 repeated trials).

The neuromuscular features extracted in section II-C led to the dual-modal sensing feature set (sEMG-US imaging feature fusion set - two dominant sEMG features and two dominant US imaging features) and uni-modal sensing feature sets (sole sEMG or sole US imaging feature set, four sEMG features or four US imaging features). To evaluate the benefits of the dual-modal sensing feature set for predicting ankle dorsiflexion movement, multiple machine learning regression models were applied, including MLR, FFNN, GPR, and SVR (see model parameters settings in supplemental section-B, SVR-L, SVR-P, and SVR-G representing SVR model with linear, polynomial, and Gaussian kernels).

For each trial in the second experimental set (11 seconds) during the seated posture and each locomotion trial (20 seconds per speed) during the walking scenario, each regression model was trained by the uni-modal and dual-modal feature sets (see the computation time in supplemental section-C). The training data of the seated posture were randomly selected from three dorsiflexion ascending periods (i.e., θ changing from 0° to MMR) out of four, and the remaining dorsiflexion ascending period was used to test the prediction performance. Similarly, for each locomotion trial, ten swing phases out of ten gait cycles were targeted, and the training data were randomly selected from six swing phases out of ten while the remaining four swing phases were used to test the prediction performance. Root mean square error (RMSE) values and R-square (R^2) values in each training and prediction step with uni-modal and dual-modal feature sets were calculated and compared across trials and participants. Shapiro-Wilk parametric hypothesis test was used to determine the normality of the corresponding RMSE values and R^2 values. According to the results of the Shapiro-Wilk test, one-way repeated-measure analysis of variance (ANOVA) or Kruskal-Wallis tests followed by a Tukey's honestly significant difference test (Tukey's HSD) was applied to evaluate if a significant difference exists among different error groups. The significant difference level was chosen as $p < 0.05$ for all statistical tests. Effect sizes were reported as η_p^2 and Cohen's d for main effects from ANOVA or Kruskal-Walls tests and pairwise comparisons from Tukey's HSD, respectively.

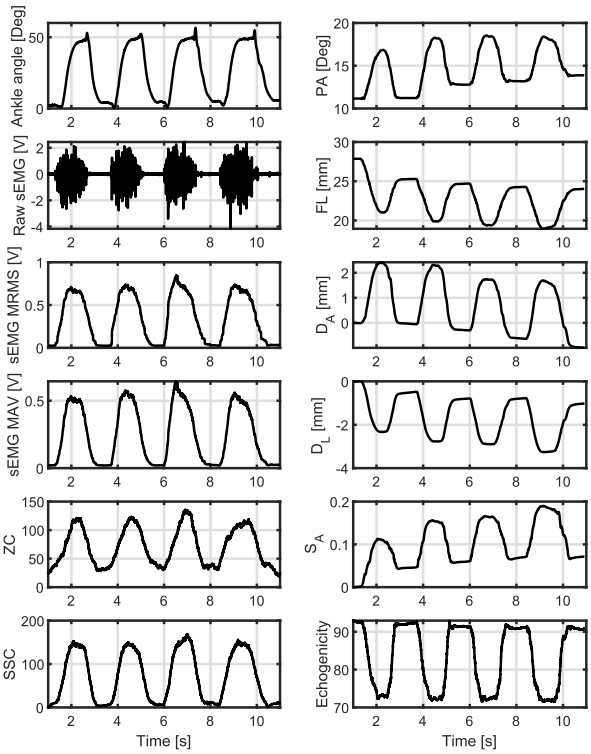


Fig. 3. Right ankle joint dorsiflexion angles and neuromuscular features from sEMG signals and US imaging in trial 3 on Participant A04 during the seated posture. MRMS - moving root mean square, MAV - mean absolute value, ZC - zero crossing, SSC - slope sign change.

III. Results and Discussions

A. Determination of Dominant Neuromuscular Features

Neuromuscular features time sequences from both sEMG signals and US imaging in a representative experimental trial during the seated posture on Participant A04 are shown in Fig. 3. The waveform length feature of sEMG signals is excluded here due to the low signal-to-noise ratio. We can easily observe that all the sEMG TD features have a positive correlation with the ankle dorsiflexion angle. For US imaging features, PA, D_A, and S_A show a positive correlation with the ankle dorsiflexion angle, while FL, D_L, and $Echo$ show a negative correlation. During the nearly periodic ankle dorsiflexion motion, the cycle-to-cycle shift of sEMG TD features is negligible, but this phase shift is obvious in US imaging features. The time-related shifts of PA and FL in Fig. 3 are related to the usage of the optical flow tracking algorithm, while the shifts of D_A, D_L, and S_A are related to the usage of the adaptive speckle tracking algorithm. Both tracking algorithms rely on the accumulation of the pixels displacement across US image time sequence. During the US imaging beamforming, it is impossible to completely remove the image noise although 2D filtering methods are employed, therefore, it is very likely the pixels tracking results are deteriorated by image noise and drifted with time. In addition, due to the elasticity property of the muscle, the pixels could be superimposed with the muscle contraction, which could also cause feature tracking drifts with time. Although currently an efficient way to eliminate the shifts of US imaging features does not exist for a online implementation, the applied machine learning models are capable of addressing the time-related shift problems. The subsequent results of model calibration and prediction do not show significant effects of the shifts on the outcomes.

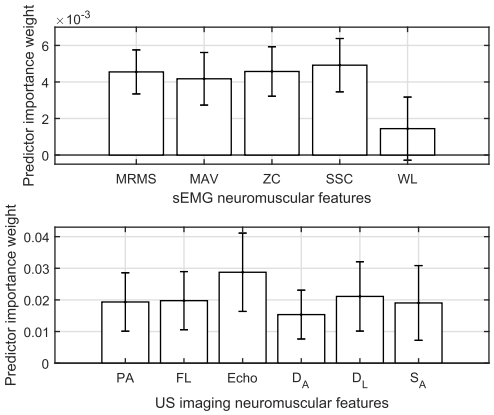
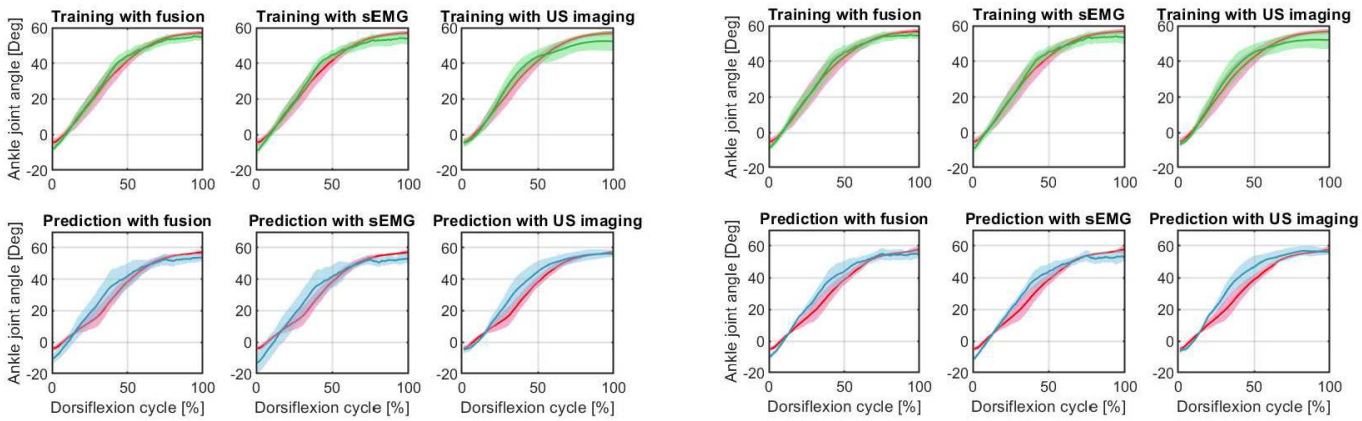


Fig. 4. The importance weight values (mean ± SD) between each sEMG-derived or US imaging-derived feature and ankle dorsiflexion angle during the seated posture across five repeated 11-second trials and all eight participants.

One possible limitation is that the time-related drifts of the US imaging-derived features might have significant effects on the outcomes. Therefore, we suggest two possible methods to eliminate the time-related drifts in future study: 1) a manual key-frame correction method [45] (post-processing), and 2) a high-pass filtering method (real-time processing or post-processing). The manual key-frame correction is challenging though due to the fact that these key-frame nodes are usually selected subjectively during the cyclic muscle activity. We are using this method in a parallel study to extract plantarflexors' thickness during the walking stance phase, where the key-frame nodes were selected as the time instants of heel-strike [37]. Since we did not have clear key-frame nodes in the seated posture setup, the manual correction was not applied for both PA and FL in this study. Our future work will address the drift problem using a filtering algorithm.

The mean value and SD of importance weights between the sEMG feature set/US imaging feature set and ankle dorsiflexion angle across repeated trials and all participants during the seated posture are shown in Fig. 4. The rank of the sEMG TD features are SSC, ZC, $MRMS$, MAV, and WL, while the rank of the US imaging features are $Echo$, D_L, FL, PA, S_A, and D_A. Therefore, the two dominant sEMG features are selected as two features with the highest importance weight values, SSC (0.0049 ± 0.0015) and ZC (0.0042 ± 0.0014), as shown in Fig. 4. Similarly, the two dominant US imaging features are selected as $Echo$ (0.0287 ± 0.0124) and D_L (0.0211 ± 0.0110). Then, three feature sets including the uni-modal sEMG feature set (four features: SSC, ZC, $MRMS$, and MAV), the uni-modal US imaging feature set (four features: $Echo$, D_L, FL, and PA), and the dual-modal sEMG-US imaging feature fusion set (four dominant features: SSC, ZC, $Echo$, and D_L) are determined and to be implemented in multiple machine learning regression models for scenarios of both the seated posture and walking at multiple speeds.



(a) SVR-L regression model (b) FFNN regression model

Fig. 5. Model training and prediction performance during dorsiflexion ascending period with uni-modal and dual-modal neuromuscular feature sets by using SVR-L and FFNN regression models on Participant A01. The dorsiflexion angle measurement, trained angles, and predicted angles are represented as red, green, and blue lines. The centered solid lines and light shadowed areas represent the mean values and standard deviation values. 60 (3 randomly selected cycles × 4 cycles each trial × 5 trials) and 20 (1 remaining cycle × 4 cycles each trial × 5 trials) dorsiflexion ascending cycles are included in training and prediction procedures, respectively.

B. Superior Performance of Dual-Modal Feature Set

Figure 5 shows the summarized training and prediction performance with the dual-modal and uni-modal sensing feature sets by using SVR-L and FFNN regression models on Participant A01 during the seated posture, where the horizontal axis represents the normalized ankle dorsiflexion ascending cycle and the vertical axis represents the ankle joint angle. Each training subplot and prediction subplot in Fig. 5 (a) and (b) consist of 60 and 20 dorsiflexion ascending cycles, respectively. For convenient notation, the RMSE between sEMG feature set-based training (or prediction) and angle measurement, between US imaging feature set-based training (or prediction) and angle measurement, and between feature fusion set-based training (or prediction) and angle measurement are denoted as $RMSE_{sEMG}^{train}$ ($RMSE_{sEMG}^{predict}$), $RMSE_{US}^{train}$ ($RMSE_{US}^{predict}$), and $RMSE_{both}^{train}$ ($RMSE_{both}^{predict}$), respectively. In Fig. 5 (a), when SVR-L model is applied, $RMSE_{sEMG}^{train} = 3.78 \pm 1.12°$, $RMSE_{US}^{train} = 4.42 \pm 1.36°$, and $RMSE_{both}^{train} = 2.55 \pm 0.75°$, respectively; while $RMSE_{sEMG}^{predict} = 4.22 \pm 1.35°$, $RMSE_{US}^{predict} = 5.41 \pm 1.83°$, and $RMSE_{both}^{predict} = 2.81 \pm 0.74°$, respectively. In Fig. 5 (b), when FFNN model is applied, $RMSE_{sEMG}^{train} = 2.97 \pm 0.87°$, $RMSE_{US}^{train} = 3.61 \pm 1.24°$, and $RMSE_{both}^{train} = 1.77 \pm 0.45°$, respectively; while $RMSE_{sEMG}^{predict} = 4.47 \pm 1.75°$, $RMSE_{US}^{predict} = 5.48 \pm 1.85°$, and $RMSE_{both}^{predict} = 2.34 \pm 0.69°$, respectively.

Furthermore, to determine the generalizability of the observed results above by using the proposed dual-modal approach during the seated posture, validations across participants and trials (VAPT) were investigated. The Shapiro-Wilk test showed that the 160 prediction RMSE values (1 remaining cycle × 4 cycles each trial × 5 trials × 8 participants) and R^2 values in VAPT by using each feature set and each regression model followed a normal distribution. Therefore, one-way repeated-measure ANOVA, followed by a Tukey's HSD, was applied to perform the statistical significance analysis. The prediction RMSE results in VAPT by using uni-modal and dual-modal feature sets and different regression models are shown in Fig. 6. The results show that the dual-modal feature set reduced the mean prediction RMSE value to less than 3° except for the GPR model. Compared to uni-modal feature sets for each regression model, the dual-modal feature set always resulted in significantly lower prediction RMSE. However, there was no significant difference for prediction RMSE values between the use of the sEMG feature set and the US imaging feature set. All the significance levels related to prediction RMSE are marked in Fig. 6.

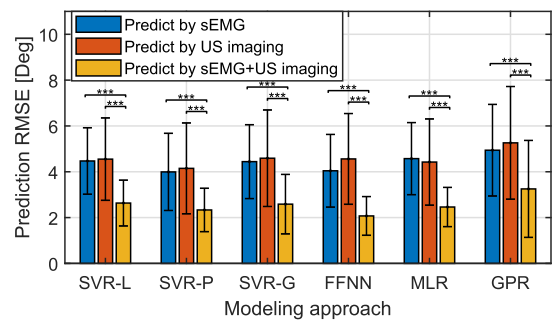


Fig. 6. Results summary of the mean ± standard deviation of ankle dorsiflexion motion prediction RMSE values by using three feature sets and six machine learning regression models in VAPT. Asterisks ⋆ represents the significant difference level of $p < 0.05$, ⋆⋆ represents the significant difference level of $p < 0.01$, and ⋆⋆⋆ represents the significant difference level of $p < 0.001$.

To quantitatively illustrate the prediction improvement performance by using the dual-modal feature set, the η_p^2 values from ANOVA, the mean prediction RMSE reduction value, reduction percentage, and significant difference level from Tukey's HSD are listed in Table I. Overall, compared to the sEMG feature set and US imaging feature set, the sEMG-US imaging feature fusion set significantly reduced the prediction RMSE from 33.25% to 47.84% and from 37.63% to 53.95%, respectively, across different regression models. Furthermore, the results show that all the R^2 values between the angular position prediction and measurement are higher than 0.8 with

either dual-modal or uni-modal feature sets for all regression models, which indicates a highly linear correlation between the prediction and measurement. Particularly, all R^2 values are higher than 0.95 with the dual-modal feature set for SVR-L, SVR-P, SVR-G, FFNN, and MLR models. Compared to the uni-modal feature sets, the dual-modal feature set significantly increased the R^2 values when SVR-L ($p < 0.001$, $\eta_p^2 = 0.13$), SVR-G ($p < 0.01$, $\eta_p^2 = 0.05$), FFNN ($p < 0.001$, $\eta_p^2 = 0.12$), and MLR ($p < 0.01$, $\eta_p^2 = 0.14$) models were applied. However, when SVR-P was applied, there was no significant difference for R^2 values between the dual-modal feature fusion set and uni-modal sEMG feature set ($p = 0.277$, $d = 0.49$), and when GPR was applied, there was no significant difference for R^2 values between the dual-modal feature fusion set and uni-modal feature sets ($p = 0.933$, $\eta_p^2 < 0.01$).

Similarly for the walking experiment results, the dual-modal feature fusion set and uni-modal feature sets were kept the same as determined in section III-A. Preliminary results from walking experiments at different speeds on two able-bodied participants are shown below to fulfill the gap between the controlled condition (seated posture scenario) above and the real application (walking scenario). FFNN was selected as the representative regression model to demonstrate the superior ankle dorsiflexion motion prediction performance during dynamic walking tasks. Fig. 7 presents the ankle joint motion prediction RMSE values during the walking swing phase at five different speeds. It is observed that the mean $RMSE_{sEMG}^{predict}$ value increases as the walking speed increases while the mean $RMSE_{US}^{predict}$ and $RMSE_{both}^{predict}$ values exhibit a quadratic trend as the walking speed increases. The prediction RMSE values are significantly affected by the feature sets at all walking speeds (main effect, $p < 0.001$ for 5 different speeds, $\eta_p^2 = 0.136, 0.036, 0.008, 0.013, 0.127$, respectively). Compared to the sEMG and US imaging feature set, the feature fusion set significantly reduced the prediction RMSE value by 47.93% ($p < 0.001$) and 27.46% ($p = 0.031$) at 0.50 m/s, by 77.66% ($p < 0.001$) and 58.39% ($p < 0.001$) at 0.75 m/s, by 74.74% ($p < 0.001$) and 29.40% ($p = 0.004$) at 1.00 m/s, by 77.72% ($p < 0.001$) and 42.58% ($p < 0.01$) at 1.25 m/s, and by 36.85% ($p < 0.001$) and 29.34% ($p < 0.01$) at 1.50 m/s, respectively.

C. Discussions

In this paper, we proposed a dual-modal approach that fused sEMG TD features and US imaging morphological/functional features to predict human volitional dorsiflexion motion through different regression models. The dynamic ankle dorsiflexion experiments were conducted on eight and two able-bodied participants during the seated posture and the walking tasks at multiple speeds, respectively. The results from these two scenarios indicated that the dual-modal approach provides a superior ankle dorsiflexion motion prediction performance, compared to uni-modal approaches.

In Fig. 6 and Fig. 7, we observed that during the seated posture the uni-modal US imaging feature set was relatively comparable to the uni-modal sEMG feature set. However, the uni-modal US imaging feature set significantly

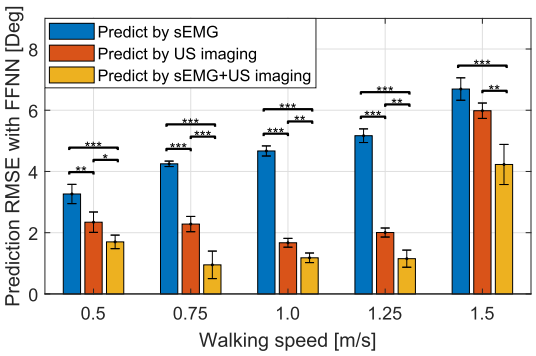


Fig. 7. RMSE mean ± standard deviation values of ankle dorsiflexion motion prediction during the walking swing phase by using three feature sets and FFNN regression model on two able-bodied participants. Each bar plot contains 8 prediction RMSE values (4 swing phases × 2 participants).

outperformed the uni-modal sEMG feature set during the walking trials. One possible reason could be that the muscle contraction level during the seated posture was much higher (i.e., dorsiflex the ankle joint as much as possible) than during the walking swing phase (i.e., purely to realize ground clearance). Given the sEMG's baseline noise was relatively steady for those two scenarios, it might have resulted in a higher sEMG's signal-to-noise ratio (SNR) during the seated posture but a lower SNR during the walking swing phase. Thus, these results also indicate that US imaging may have a more reliable performance across different dynamic scenarios.

sEMG signals reflect the excitation superposition of motor units in the targeted muscle area, which precedes the corresponding joint motion. In the present study, we assessed the feasibility that US imaging features of the TA muscle also precede the ankle joint motion. As shown in Fig. 2, the observed US imaging preceding time gap was from 45 ms to 85 ms. These observations agree with a study by Begovic *et al.* [46] who reported that the time delay between onset of muscle fiber motion and the muscle contraction force generation measured by US signal is 49.7 ± 7.0 ms during volitional contraction of quadriceps femoris muscles. Considering the delay between the onset of muscle force generation and joint motion, the observed US imaging preceding time gap seems like a realistic estimate for the volitional joint motion prediction. Existing literature has successfully shown the effectiveness of using uni-modal US imaging features to estimate or predict joint motion. For example, Jahanandish *et al.* [29] extracted five US features of the rectus femoris muscle based on a multi-scale ridge filter and a random sample consensus model and applied the GPR model to train the features and estimate both knee joint angle and angular velocity. Sikdar *et al.* [47] used the nearest neighbor classifier to create mappings between changes in the US echogenicity of forearm muscles and finger movements. In their recent work [23], Pearson's correlation coefficient was used between the *rest* frame and other US frames in the motion sequence to classify volitional motion intent. Our motivation in the current study is to develop a dual-modal approach that fuses sEMG and US imaging and predicts continuous volitional ankle joint motion. Compared to the uni-modal feature set, the overall improvement of the prediction RMSE with the dual-modal feature set is from 33.25%

TABLE I
PREDICTION RMSE REDUCTION WITH THE NEUROMUSCULAR FEATURE FUSION SET COMPARED TO SOLE sEMG FEATURE SET OR US IMAGING FEATURE SET IN VAPT

$RMSE_{both}^{predict}$ reduction	SVR-L	SVR-P	SVR-G	FFNN	MLR	GPR
ANOVA η_p^2	0.28	0.23	0.21	0.35	0.31	0.16
Compared to $RMSE_{sEMG}^{predict}$	1.87° (42.18%)	1.69° (41.39%)	1.89° (42.78%)	1.93° (47.84%)	2.01° (45.19%)	1.65° (33.25%)
Tukey's HSD $p-$value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Tukey's HSD Cohen's d	-1.48	-1.25	-1.29	-1.57	-1.63	-0.83
Compared to $RMSE_{US}^{predict}$	1.95° (42.66%)	1.85° (43.85%)	2.01° (43.90%)	2.44° (53.95%)	1.98° (44.77%)	1.92° (37.63%)
Tukey's HSD $p-$value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Tukey's HSD Cohen's d	-1.35	-1.15	-1.14	-1.64	-1.37	-0.89

to 47.84% and from 37.63% to 53.95%, respectively, under the seated posture scenario. As indicated in Table I, although the dual-modal approach resulted in the prediction RMSE reduction ranging from 1.65° up to 2.44° when compared to the uni-modal approach, the improvement may significantly aid toe clearance during the swing phase. The toe clearance is generally very close to the walking surface, approximately 10 mm between the foot and ground during normal gait [48]. Without considering the hip flexion's contribution to the toe clearance, a small ankle joint motion error (for example 2°) is very likely to cause tripping risk and even fall risk. Therefore, the depicted improvement in ankle dorsiflexion prediction accuracy in this paper may enable effective toe clearance with rehabilitative devices.

The nearly cyclic volitional joint dorsiflexion prediction results in this study are comparable to work done by other research groups. Usually, the joint motion prediction is considered excellent if the percentage of RMSE normalized to peak joint angle is smaller than 15% [49]. As shown in Fig. 6 (a), the percentage values of mean prediction RMSE normalized to the lowest individual MMR are all less than 14.02% with uni-modal feature sets and less than 8.76% with the dual-modal feature set, which proves all prediction results are excellent. The RMSE values of EMG-driven model in [50] were between 15.98° and 36.06° for multiple full cycles of elbow joint motion with the peak joint position around 120°. The RMSE values of the EMG-driven model in [30] for single elbow flexion and single elbow extension movements were 34.64° $\pm$ 7.79° and 18.67° $\pm$ 8.49° with the motion range between 0° and 140°. Ha *et al.* [51] were able to predict the sound knee joint with RMSE values of 5.2° $\pm$ 1.0° with the cyclic motion range between $-30°$ and $-70°$. Compared to the above EMG-driven neuromuscular models for joint motion prediction, the prediction RMSE values of the ankle dorsiflexion movement with the proposed dual-modal approach demonstrated satisfactory performance.

While the proposed dual-modal approach provides promising potentials for improving volitional ankle dorsiflexion motion prediction accuracy, we also found some critical remarks. For example, the comparison of the regression accuracy is indeed related to the quality of the sensors used in the study. The noise rating of the applied sEMG sensor is 1.2 μV (RMS, R.T.I), the sEMG signal collection system is $\leq$ 1.2 μV (RMS, R.T.I), and the US transducer receiver is <1 $nV/\sqrt{Hz}$ (see more products specifications in supplemental section-D). The optical flow algorithms proposed in [41], [42] were applied to extract routine structural features PA and FL from each single US imaging frame, where two ending points of the targeted muscle fascicle need to be defined in the initial image frame, and the US image sequence with high quality and highly visualized muscle fascicles is preferable for a reliable tracking. However, for dynamic joint motions, it is challenging to guarantee that the visualized fascicles are always well imaged due to the US transducer orientation with respect to muscle longitudinal direction, the elevation angle between the transducer and skin, and the slippery between transducer and skin [52]. For functional US imaging features, there is no need to define specific pixels' locations in the initial frame, which is less dependent on clearly visualized fascicles. Among the functional features, echogenicity is an easily accessible measurement derived from US imaging that does not require complex tracking algorithms, thus saving image processing time and potentially enabling real-time implementation with minimal effort. In addition, low image quality and low SNR may cause the problem of tissue displacement and strain features that the speckle tracking algorithm is tracking the noise instead of the useful pixel information. Therefore, a good noise filtering method is necessary before implementing a speckle tracking algorithm for the US imaging sequence. Although the aforementioned explicit sEMG or US imaging features have been well established and accepted in the literature, one limitation exists that they significantly reduce the information dimension of the original signals and neglect implicit deep features. The proposed dual-modal approach could provide complementary information of the muscle contraction from both electrical and mechanical aspects that uni-modal approaches cannot handle. In future work, to address the significant reduction of the information dimension, one potential research direction would be exploring deep features either from uni-model sEMG or US imaging signals, like deep convolutional neural network (CNN), which could generate comparable or superior joint motion prediction performance when compared to the uni-modal approach in the current study. For example, we briefly investigated a deep CNN for ankle dorsiflexion movement prediction during the walking swing phase, and the preliminary results (see details in supplemental section-E) showed that the deep CNN was a promising approach by using the uni-modal US imaging signal and without extracting any explicit US imaging features.

In the current study, the controlled seated condition was used to imitate the scenario of dynamic ankle dorsiflexion during the walking swing phase. The only difference between the seated condition and the walking condition is the knee joint configuration, $\sim$90° for the seated condition while 0°$-\sim$ 20°

for the walking condition. The preliminary walking experimental results filled the gap between the research seated conditions and the real application. However, there are some limitations in the current study. First, the preliminary dynamic walking experimental results were only from two able-bodied participants, and further experiments on more participants are needed to generalize the findings for dynamic walking scenarios in the current study. Then, the current study only focused on the agonist TA muscle to predict ankle dorsiflexion motion, and the effects of the antagonist's muscles, like gastrocnemius and soleus muscles, have not yet been considered due to 1) the limited availability of US transducers in our lab and 2) space restrictions on the shank that limit the placement of multiple US transducers. Future work will also include the new design of a US transducer holder that can simultaneously enable the measures of multiple muscles, including TA, gastrocnemius, and soleus muscles.

Lastly, the current results indeed showed that the proposed dual-modal approach worked well for healthy participants. However, it is unclear whether the feature fusion between sEMG and US imaging can better predict the motor output when pathological muscles are involved, where abnormal muscle activation levels in the clinical population are very common. The next step for this study will focus on testing the feasibility of the dual-modal approach during the swing phase of walking conditions on a customized treadmill for both more able-bodied participants and patients with incomplete spinal cord injury (iSCI). The walking speed will be set the same in the current study for able-bodied participants, and be set as the self-selected one for patients with iSCI. The data collection will include TA's sEMG and US imaging signals, ankle joint kinematics signals, and ground reaction force signals. All continuously collected data will be segmented to the individual swing phase for further analysis by using the proposed dual-modal approach.

IV. Conclusion

The present study investigated the feasibility of a dual-modal approach by fusing TA muscle neuromuscular features from both sEMG signal and US B-mode imaging for volitional ankle dorsiflexion movement prediction in both seated posture and walking at multiple speeds. The ankle joint motion prediction performance of the proposed dual-modal approach was evaluated through multiple regression models. Dominant features from synchronously collected sEMG signal and US imaging were extracted based on the rank of importance weights between the sEMG feature set/US imaging feature set and ankle dorsiflexion angle. The seated posture experimental results on eight able-bodied participants showed that overall the dual-modal feature set significantly reduced prediction RMSE from 33.25% to 47.84%, and from 37.63% to 53.95% across different regression models, compared to sole sEMG feature set and US imaging feature set, respectively. The mean prediction RMSE value was reduced to less than 9% (normalized to the corresponding MMR) by using the dual-modal feature set for each applied regression model. Furthermore, the effectiveness of the dual-modal approach was also validated in the dynamic walking scenarios. The precise motion prediction results motivate the potential implementation toward an sEMG-US imaging-based sensing interface for volitional control of assistive devices. In addition, the reliable prediction of joint motion will contribute to improvements of the intuitive and seamless human-robot integration, motion intent detection, and more advanced assist-as-needed control strategies.

References

[1] G. M. Lyons, T. Sinkjær, J. H. Burridge, and D. J. Wilcox, "A review of portable FES-based neural orthoses for the correction of drop foot," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 10, no. 4, pp. 260–279, Dec. 2002.

[2] G. Chen, C. K. Chan, Z. Guo, and H. Yu, "A review of lower extremity assistive robotic exoskeletons in rehabilitation therapy," *Crit. Rev. Biomed. Eng.*, vol. 41, nos. 4–5, pp. 343–363, 2013.

[3] A. J. Young and D. P. Ferris, "State of the art and future directions for lower limb robotic exoskeletons," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 2, pp. 171–182, Feb. 2017.

[4] S. K. Sabut, C. Sikdar, R. Kumar, and M. Mahadevappa, "Functional electrical stimulation of dorsiflexor muscle: Effects on dorsiflexor strength, plantarflexor spasticity, and motor recovery in stroke patients," *Neurorehabilitation*, vol. 29, no. 4, pp. 393–400, 2011.

[5] T. M. Kesar *et al.*, "Functional electrical stimulation of ankle plantarflexor and dorsiflexor muscles: Effects on poststroke gait," *Stroke*, vol. 40, no. 12, pp. 3821–3827, Dec. 2009.

[6] S. Srivastava *et al.*, "Assist-as-needed robot-aided gait training improves walking function in individuals following stroke," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 23, no. 6, pp. 956–963, Nov. 2015.

[7] S. Hussain, P. K. Jamwal, M. H. Ghayesh, and S. Q. Xie, "Assist-as-needed control of an intrinsically compliant robotic gait training orthosis," *IEEE Trans. Ind. Electron.*, vol. 64, no. 2, pp. 1675–1685, Feb. 2017.

[8] S. Wang *et al.*, "Design and control of the MINDWALKER exoskeleton," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 23, no. 2, pp. 277–286, Mar. 2015.

[9] D. Zanotto, Y. Akiyama, P. Stegall, and S. K. Agrawal, "Knee joint misalignment in exoskeletons for the lower extremities: Effects on user's gait," *IEEE Trans. Robot.*, vol. 31, no. 4, pp. 978–987, Aug. 2015.

[10] Q. Zhang, K. Kim, and N. Sharma, "Prediction of ankle dorsiflexion moment by combined ultrasound sonography and electromyography," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 1, pp. 318–327, Jan. 2020.

[11] Q. Zhang, A. Iyer, K. Kim, and N. Sharma, "Evaluation of non-invasive ankle joint effort prediction methods for use in neurorehabilitation using electromyography and ultrasound imaging," *IEEE Trans. Biomed. Eng.*, vol. 68, no. 3, pp. 1044–1055, Mar. 2021.

[12] T. Matsubara and J. Morimoto, "Bilinear modeling of EMG signals to extract user-independent features for multiuser myoelectric interface," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 8, pp. 2205–2213, Aug. 2013.

[13] N. Bu, M. Okamoto, and T. Tsuji, "A hybrid motion classification approach for EMG-based human–robot interfaces using Bayesian and neural networks," *IEEE Trans. Robot.*, vol. 25, no. 3, pp. 502–511, Jun. 2009.

[14] Y. Huang, K. B. Englehart, B. Hudgins, and A. D. C. Chan, "A Gaussian mixture model based classification scheme for myoelectric control of powered upper limb prostheses," *IEEE Trans. Biomed. Eng.*, vol. 52, no. 11, pp. 1801–1811, Nov. 2005.

[15] M. Khezri and M. Jahed, "A neuro–fuzzy inference system for sEMG-based identification of hand motion commands," *IEEE Trans. Ind. Electron.*, vol. 58, no. 5, pp. 1952–1960, May 2011.

[16] M. Sartori, D. Farina, and D. G. Lloyd, "Hybrid neuromusculoskeletal modeling to best track joint moments using a balance between muscle excitations derived from electromyograms and optimization," *J. Biomech.*, vol. 47, no. 15, pp. 3613–3621, 2014.

[17] G. Durandau, D. Farina, and M. Sartori, "Robust real-time musculoskeletal modeling driven by electromyograms," *IEEE Trans. Biomed. Eng.*, vol. 65, no. 3, pp. 556–564, Mar. 2018.

[18] Q. L. Li, Y. Song, and Z. G. Hou, "Estimation of lower limb periodic motions from sEMG using least squares support vector regression," *Neural Process. Lett.*, vol. 41, no. 3, pp. 371–388, 2015.

[19] F. Zhang *et al.*, "sEMG-based continuous estimation of joint angles of human legs by using BP neural network," *Neurocomputing*, vol. 78, no. 1, pp. 139–148, 2012.

[20] L. J. Hargrove, K. Englehart, and B. Hudgins, "A comparison of surface and intramuscular myoelectric signal classification," *IEEE Trans. Biomed. Eng.*, vol. 54, no. 5, pp. 847–853, May 2007.

[21] D. L. Crouch, L. Pan, W. Filer, J. W. Stallings, and H. Huang, "Comparing surface and intramuscular electromyography for simultaneous and proportional control based on a musculoskeletal model: A pilot study," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 9, pp. 1735–1744, Sep. 2018.

[22] N. Akhlaghi *et al.*, "Real-time classification of hand motions using ultrasound imaging of forearm muscles," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 8, pp. 1687–1698, Aug. 2016.

[23] A. S. Dhawan *et al.*, "Proprioceptive sonomyographic control: A novel method for intuitive and proportional control of multiple degrees-of-freedom for individuals with upper extremity limb loss," *Sci. Rep.*, vol. 9, no. 1, pp. 1–15, Dec. 2019.

[24] J. He, H. Luo, J. Jia, J. T. W. Yeow, and N. Jiang, "Wrist and finger gesture recognition with single-element ultrasound signals: A comparison with single-channel surface electromyogram," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 5, pp. 1277–1284, May 2019.

[25] X. Yang, X. Sun, D. Zhou, Y. Li, and H. Liu, "Towards wearable a-mode ultrasound sensing for real-time finger motion recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 6, pp. 1199–1208, Jun. 2018.

[26] J.-Y. Guo, Y.-P. Zheng, H.-B. Xie, and X. Chen, "Continuous monitoring of electromyography (EMG), mechanomyography (MMG), sonomyography (SMG) and torque output during ramp and step isometric contractions," *Med. Eng. Phys.*, vol. 32, no. 9, pp. 1032–1042, Nov. 2010.

[27] E. M. Strasser, T. Draskovits, M. Praschak, M. Quittan, and A. Graf, "Association between ultrasound measurements of muscle thickness, pennation angle, echogenicity and skeletal muscle strength in the elderly," *Age*, vol. 35, no. 6, pp. 2377–2388, Dec. 2013.

[28] R. W. Nuckols, K. Swaminathan, S. Lee, L. Awad, C. J. Walsh, and R. D. Howe, "Automated detection of soleus concentric contraction in variable gait conditions for improved exosuit control," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2020, pp. 4855–4862.

[29] M. H. Jahanandish, N. P. Fey, and K. Hoyt, "Lower limb motion estimation using ultrasound imaging: A framework for assistive device control," *IEEE J. Biomed. Health Inform.*, vol. 23, no. 6, pp. 2505–2514, Nov. 2019.

[30] T. K. K. Koo and A. F. T. Mak, "Feasibility of using EMG driven neuromusculoskeletal model for prediction of dynamic movement of the elbow," *J. Electromyogr. Kinesiol.*, vol. 15, no. 1, pp. 12–26, Feb. 2005.

[31] I. Batzianoulis, N. E. Krausz, A. M. Simon, L. Hargrove, and A. Billard, "Decoding the grasping intention from electromyography during reaching motions," *J. NeuroEng. Rehabil.*, vol. 15, no. 1, p. 57, 2018.

[32] M. H. Jahanandish, K. G. Rabe, N. P. Fey, and K. Hoyt, "Ultrasound features of skeletal muscle can predict kinematics of upcoming lower-limb motion," *Ann. Biomed. Eng.*, vol. 49, no. 2, pp. 822–833, Feb. 2021.

[33] W. Niu, Y. Wang, Y. He, Y. Fan, and Q. Zhao, "Kinematics, kinetics, and electromyogram of ankle during drop landing: A comparison between dominant and non-dominant limb," *Hum. Movement Sci.*, vol. 30, no. 3, pp. 614–623, Jun. 2011.

[34] R. J. Cunningham, P. J. Harding, and I. D. Loram, "The application of deep convolutional neural networks to ultrasound for modelling of dynamic states within human skeletal muscle," 2017, *arXiv:1706.09450*. [Online]. Available: http://arxiv.org/abs/1706.09450

[35] Y. Fang, N. Hettiarachchi, D. Zhou, and H. Liu, "Multi-modal sensing techniques for interfacing hand prostheses: A review," *IEEE Sensors J.*, vol. 15, no. 11, pp. 6065–6076, Nov. 2015.

[36] M. Robnik-Šikonja and I. Kononenko, "Theoretical and empirical analysis of ReliefF and RReliefF," *Mach. Learn.*, vol. 53, no. 1, pp. 23–69, 2003.

[37] Q. Zhang, N. Fragnito, A. Myers, and N. Sharma, "Plantarflexion moment prediction during the walking stance phase with an sEMG-ultrasound imaging-driven model," in *Proc. EMBC Conf. Paper*, Nov. 2021.

[38] B. Hudgins, P. Parker, and R. N. Scott, "A new strategy for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 40, no. 1, pp. 82–94, Jan. 1993.

[39] O. W. Samuel *et al.*, "Pattern recognition of electromyography signals based on novel time domain features for amputees' limb motion classification," *Comput. Electr. Eng.*, vol. 67, pp. 646–655, Apr. 2018.

[40] A. Georgakis, L. K. Stergioulas, and G. Giakas, "Fatigue analysis of the surface EMG signal in isometric constant force contractions using the averaged instantaneous frequency," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 2, pp. 262–265, Feb. 2003.

[41] N. J. Cronin, C. P. Carty, R. S. Barrett, and G. Lichtwark, "Automatic tracking of medial gastrocnemius fascicle length during human locomotion," *J. Appl. Physiol.*, vol. 111, no. 5, pp. 1491–1496, Nov. 2011.

[42] J. G. Gillett, R. S. Barrett, and G. A. Lichtwark, "Reliability and accuracy of an automated tracking algorithm to measure controlled passive and active muscle fascicle length changes from ultrasound," *Comput. Methods Biomech. Biomed. Eng.*, vol. 16, no. 6, pp. 678–687, Jun. 2013.

[43] Z. Sheng, N. Sharma, and K. Kim, "Quantitative assessment of changes in muscle contractility due to fatigue during NMES: An ultrasound imaging approach," *IEEE Trans. Biomed. Eng.*, vol. 67, no. 3, pp. 832–841, Mar. 2020.

[44] Z. Sheng, N. Sharma, and K. Kim, "Ultra-high-frame-rate ultrasound monitoring of muscle contractility changes due to neuromuscular electrical stimulation," *Ann. Biomed. Eng.*, vol. 49, pp. 262–275, Jan. 2021.

[45] D. J. Farris and G. A. Lichtwark, "UltraTrack: Software for semi-automated tracking of muscle fascicles in sequences of B-mode ultrasound images," *Comput. Methods Programs Biomed.*, vol. 128, pp. 111–118, May 2016.

[46] H. Begovic, G.-Q. Zhou, T. Li, Y. Wang, and Y.-P. Zheng, "Detection of the electromechanical delay and its components during voluntary isometric contraction of the quadriceps femoris muscle," *Frontiers Physiol.*, vol. 5, p. 494, Dec. 2014.

[47] S. Sikdar *et al.*, "Novel method for predicting dexterous individual finger movements by imaging muscle activity using a wearable ultrasonic system," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 1, pp. 69–76, Jan. 2014.

[48] D. Winter, *Biomechanics and Motor Control of Human Movement*. Hoboken, NJ, USA: Wiley, 2009.

[49] M. M. Liu, W. Herzog, and H. H. C. M. Savelberg, "Dynamic muscle force predictions from EMG: An artificial neural network approach," *J. Electromyogr. Kinesiol.*, vol. 9, no. 6, pp. 391–400, Dec. 1999.

[50] J. W. L. Pau, S. S. Q. Xie, and A. J. Pullan, "Neuromuscular interfacing: Establishing an EMG-driven model for the human elbow joint," *IEEE Trans. Biomed. Eng.*, vol. 59, no. 9, pp. 2586–2593, Sep. 2012.

[51] K. H. Ha, H. A. Varol, and M. Goldfarb, "Volitional control of a prosthetic knee using surface electromyography," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 1, pp. 144–151, Jan. 2011.

[52] A. Boezaart and B. Ihnatsenka, "Ultrasound: Basic understanding and learning the language," *Int. J. Shoulder Surg.*, vol. 4, no. 3, p. 55, 2010.